

16/08/2026

Network Device

① HUB.

- It is a layer 1 device and half duplex
- It does broadcast to all connected devices
- If A PC sends a signal to B PC, this signal will go to C PC as well and C PC will read the mac address and drop the signal. B PC will respond to A PC.
- It creates signal collision & data traffic everytime

② Explain SWITCH.

- It is a layer 2 device (Data link layer)
- When A PC sends data through switch - switch will do address learning from the mac table.
- If the destination PC mac is stored it will send the data to the destination PC and does Unicast.

→ if destination mac is not stored in MAC table. Switch will do broadcast and other PC's will drop if D.Mac is not matched.

→ MAC table timer limit is 5mins

→ first time ARP will be sent from source PC to switch.

→ Reduces traffic

→ It is a single broadcast domain

→ It should be in same network

③ Router:

→ It is layer 3 [Network]

→ devices should be connected in different network.

→ Routing table consists of (Network ID, prefix, int name)

→ It is used to breakdown broadcast domain

→ IP config is mandatory.

→ 1st IP is always given to router interface

→ Data will be route according to the network.